

Johnathon Beuter

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Education

Iowa State University of Science and Technology, Ames IA

- Master of Science in Cyber Security GPA: 3.77 January 2025 – May 2026
- Bachelor of Science in Cyber Security Engineering GPA: 3.21 August 2021 – May 2025
- Minor in Spanish

SANS Institute, Bethesda MD

- GIAC GFACT Certification December 2023 – May 2024

Professional Experience

Associate Software Engineer

Wellmark Blue Cross and Blue Shield, Des Moines IA | August 2026 - Present

- Identified, documented, and upgraded vulnerable Node.js dependencies, using Github dependabot, reducing security risk across team code repositories
- Developed Copilot AI agents to automate repetitive task within Azure DevOps, improving team efficiency and reducing manual inputs to teams' Agile processes
- Wrote SQL queries to gather statistics on member data that assisted in developing new features within Wellmark's case management application

Cyber Security Teaching Assistant

Iowa State University of Science and Technology, Ames IA | August 2023 – May 2026

- Created a bi-weekly cybersecurity blog, averaging 1,400 impressions per post, covering Windows exploitation, network attacks, and other offensive security topics (see URL above for recent posts)
- Authored new course content focusing on lateral movement, using SSH tunnels and Proxychains, and Java deserialization exploits
- Revised 12+ lab exercises on lin/winPEAS, Bash scripting, nmap, and reverse shells, updating content and improving instructional quality

Penetration Tester

IP Pathways, Urbandale IA | January 2026 – May 2026

- Conducted internal penetration tests for regional credit unions, school districts, and nonprofit organizations, achieving Domain Admin (or root account access) on 60% of engagements
- Designed Bash and Python scripts to automate scanning, organize tool output, and chain together exploits to enhance exploitation capabilities
- Independently authored penetration test reports containing executive summaries, technical findings, risk ratings, proof-of-concept evidence, and remediation guidance for technical and non-technical stakeholders

Cyber Security Engineering Intern

Wellmark Blue Cross and Blue Shield, Des Moines IA | May 2025 – August 2025

- Built an isolated sandbox environment using CAPEv2, Python, and Virtual Box to automate malware analysis processes used by security engineers

- Identified misconfiguration in Wellmark's use of Microsoft Configuration Manager (SCCM) and created a pen test plan using SharpSCCM and ntlmrelayx to verify vulnerabilities
- Performed daily verification of threats detected on Palo Alto firewalls and Windows Defender EDR

Red Team Intern

United States Air Force Research Laboratory, Rome NY | May 2024 - August 2024

- Engineered offensive security tools using Python and GO, capable of performing Modbus denial of services attacks against Raspberry Pis simulating Operational Technology (OT) devices
- Led weekly red team operations in Windows and Linux environments, leveraging custom built tools and Sliver C2 to conduct lateral movement, establish persistence, and evade detection
- Reverse-engineered malware samples using Ghidra and x64dbg, producing detailed technical documentation of findings

Activities

Engineers Without Borders (EWB) Volunteer

- *Iowa President | Jan. 2025 –Present:*
 - Led organizational improvement informed by collaboration with other EWB chapters and industry professionals
- *Vice President of Engineering | Jan. 2024 – May 2026:*
 - Led engineering teams to Ghana in winter of 2025 and 2026, coordinating and managing the completion of numerous construction projects with the help of regional and local stakeholders

Awards and Accolades

Teaching Excellence Award - Iowa State University

- Recognized for being within the top 10% of graduate students involved in teaching each academic year

Member of the Year 2025 - Iowa State EWB

- Recognized for outstanding membership in the Iowa State Engineers Without Borders chapter, contributing to project planning, overseas work, and community outreach initiatives that advanced organizational goals

ACE Class Valedictorian and Distinguished Graduate

- Graduated as Valedictorian of a 30-person cyber warfare training program (ACE), comprised of college students, military personnel, and civilian contractors from government agencies and top universities